Language Models Encode Numbers Using Digit Representations in Base 10

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Abstract

Large language models (LLMs) frequently make errors when handling even simple numerical problems, such as comparing two small numbers. A natural hypothesis is that these errors stem from how LLMs represent numbers, and specifically, whether their representations of numbers capture their numeric values. We tackle this question from the observation that LLM errors on numerical tasks are often distributed across *the digits* of the answer rather than normally around *its numeric value*. Through a series of probing experiments and causal interventions, we show that LLMs internally represent numbers with individual circular representations per-digit in base 10. This digit-wise representation, as opposed to a value representation, sheds light on the error patterns of models on tasks involving numerical reasoning and could serve as a basis for future studies on analyzing numerical mechanisms in LLMs.

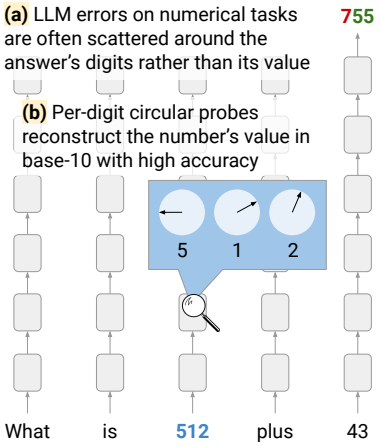

Figure 1: An illustration of our key findings, suggesting that LLMs represent numbers on a per-digit base-10 basis: (a) on simple numerical tasks, LLMs often make errors that are close to the answer in 'digit space' rather than in value space, (b) though probing the exact number is hard, digit values can be decoded accurately.

1 Introduction

Despite their high performance on various challenging tasks (Bubeck et al., 2023; Bommasani et al., 2021; Trinh et al., 2024), large language models (LLMs) often struggle with simple numerical problems, such as adding or comparing the magnitude of two small numbers. While previous works commonly attribute such failures to different limitations in the representations of LLMs (e.g., McLeish et al., 2024; Nogueira et al., 2021), *how* LLMs represent numbers is still an outstanding question.

Recently, Zhu et al. (2024) used linear probes to predict the number encoded in a hidden representation, showing high correlation with the expected value. However, the probes exhibited low accuracy, suggesting that a linear representation alone is not sufficient to explain how LLMs can often perform exact numerical operations, such as addition and multiplication. Maltoni and Ferrara (2024) have suggested that LLMs may do arithmetic in "value space", but then we would expect to see a normally-distributed error pattern, which we will see is not the case in widely-used models.

We approach the above question by observing that when models make numerical errors, the errors are often distant from the correct answer in value space but close in 'digit space'. For example, consider the simple addition problem "$132 + 238 + 324 + 139 =$ " where the correct answer is 833. LLMs are more likely to generate errors with high string-similarity to the correct answer, such as "633" or "823", than natural errors like "831" or "834", which are close in value, as if the model's internal algorithm misreads one of the digits in the input. We show this rigorously in §2.

We argue that such scattered error distributions are unlikely to occur in models that directly manipulate numbers in a value space. For example, in multi-operand addition, if the model represents

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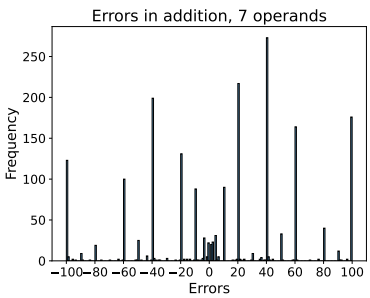

Figure 2: Error distribution in 7 operand addition.

each number in a value space and then translates the result back to tokens after addition, we would expect a normal error distribution around the correct answer. This distribution would arise from noise in the addition operation and the representations themselves. However, the observed scattered errors (in §2) suggest that the model may represent numbers in a fragmented manner, for example based on their individual digits.

To test this hypothesis, we first train probes to recover the number value and digit values from hidden representations of numbers. Our experiments with Llama 3 8B (Dubey et al., 2024) and Mistral 7B (Jiang et al., 2023) show that, while probes fail to recover the exact number value directly (which agrees with Zhu et al., 2024), the hidden representations of a number contain an orthogonal circular representation for each digit in base 10 (as illustrated in Figure 1). This observation holds across both models, which use different tokenization schemes for numbers. Moreover, causally intervening on these circular digit representations (i.e., performing $+5$ mod10) often modifies the value of the number accurately.

To conclude, our work proposes that the scattered errors LLMs demonstrate on arithmetic problems stem from a fragmented digit-wise representation of numbers. We show that this hypothesis holds in practice; it is possible to accurately recover and modify the digit values from number representations in base 10, but not the number values. Our findings provide a basis for understanding mathematical operations in LLMs and mitigating numerical errors. We release our code at `https://github.com/amitlevy/base10`.

2 Model Errors on Numerical Tasks are Scattered Across Digits

We analyze the distribution of errors by Llama 3 8B on two simple numerical tasks with numbers within the range 0 to 999, which the model represents as individual tokens. We find that errors are distributed in a digit-wise manner, where an incorrect prediction is close to the correct answer in string edit distance but not in value space.

Digit	Correct	Incorrect
Units	4,232 94%	259 6%
Tens	4,351 97%	140 3%
Hundreds	4,054 92%	356 8%

Table 1: Accuracy of Llama 3 8B in comparing the magnitude of two numbers differing by one digit.

Task 1: Multi-operand addition We generated 5,000 queries of addition of $N = 7$ operands, which sum into a number between 0 and 1000, and calculated the errors of the model on these queries. Figure 2 displays the error distribution, showing that most errors are exact multiples of 10 and 100. Further, when considering the error distributions for any number of operands between 4 and 8, we observe that about 80% of the errors are in a single output digit, which is often not the units digit. A similar error analysis of GPT-4o (OpenAI et al., 2024) on 20-operand addition tasks showed similar trends (§A.1).

Task 2: Comparison of two numbers We consider all pairs of numbers between 0 and 999, which differ from each other in only a single digit—the units, tens, or hundreds place. Given a pair of numbers, the model needs to indicate which number is larger, e.g. *"between 121 or 171, the larger number is:"*. Table 1 shows that errors are distributed approximately equally between the digits. This indicates that the model's likelihood of making a mistake is not significantly affected by the numerical closeness of the numbers, as would be expected if numbers were represented in value space.

The evident base-10 digit-related error trends in both tasks lead to the hypothesis that LLMs may represent numbers in base 10 as opposed to in a linear value space, which we test in the next section.

3 LLMs Represent Numbers Digit-Wise in Base 10

We test our hypothesis and show that LLMs represent numbers on a per-digit base 10 basis.

3.1 Experimental setting

Probing We train digit-wise probes that estimate the value of a number from its hidden representa-

Basis	2	3	4	5	6	7	8	9	10	11	12	13	14	1000	2000
Llama 3 8B	0.16	0.06	0.16	0.67	0.05	0.08	0.06	0.07	**0.91**	0.08	0.06	0.06	0.06	0.00	0.00
Mistral 7B	0.13	0.02	0.13	0.72	0.02	0.05	0.05	0.08	**0.92**	0.12	0.04	0.06	0.05	0.01	0.00

Table 2: Accuracy in predicting all digits of digit-wise circular probes in various bases, averaged over layers ≥ 3.

tion by predicting the numeric values of its digits. Let M be a pre-trained transformer-based language model (Vaswani et al., 2017) with L layers and a hidden dimension d, and denote by $\mathbf{h}_j^\ell$ the hidden representation of the j-th input token at layer ℓ. In the following, we omit the position index and use $\mathbf{h}^\ell$, as in our experiments we always consider the last position of the input (i.e., the last numeric token). For a digit i, a base b, and a layer $\ell \in [L]$, we train a circular probe (Engels et al., 2024) that given the hidden representation $\mathbf{h}^\ell$ of a number x, predicts the numeric value of its i-th digit in base b:

$$\mathbf{P}_{i,b}^\ell = \underset{\mathbf{P}' \in \mathbb{R}^{2\times d}}{\arg\min} \sum_{\langle \mathbf{h}^\ell, x_i \rangle \in \mathcal{D}^\ell} \left\| \mathbf{P}'\mathbf{h}^\ell - \texttt{circle}_b(x_i) \right\|_2^2 \tag{1}$$

$\mathcal{D}^\ell$ is a training set consisting of pairs $\langle \mathbf{h}^\ell, x_i \rangle$ of the ℓ-th layer hidden representation and the i-th digit of a number x, and

$$\texttt{circle}_b(t) = [\cos(2\pi t/b), \sin(2\pi t/b)] \tag{2}$$

maps a digit in base b to a point on the unit circle.

Using the set of probes for some layer ℓ, we define a function that reconstructs the value of a number x from its representation $\mathbf{h}^\ell$. For every digit i, define a function $\texttt{digit}_{i,b}^\ell : \mathbb{R}^d \to [b]$ that predicts the value of that digit by applying $\texttt{digit}_{i,b}^\ell := \frac{b}{2\pi} \cdot \text{atan2}(\mathbf{P}_{i,b}^\ell \mathbf{h}^\ell)$.[1] Concatenating the outputs of the functions for all the digits of x provides an estimation of its value in base b. For example, the value of a 3-digit number would be reconstructed in base b from its ℓ-layer representation by concatenating $[\texttt{digit}_{3,b}^\ell, \texttt{digit}_{2,b}^\ell, \texttt{digit}_{1,b}^\ell]$.

In addition to the circular probes, we trained linear probes, which have been used recently to extract various features from LLM representations (Belinkov, 2022; Park et al., 2023; Gurnee and Tegmark, 2024). While the linear probes showed similar trends to the circular probes, we observed they are less effective in predicting numerical values from LLM representations. This observation agrees with recent findings that some features in LLMs have non-linear representations (Engels et al., 2024) as well as with the circular patterns observed in PCA plots (see §A.2). Therefore, in our experiments we focus on circular probes.

Data For each positive number $x \in [2000]$ we feed "$\langle x \rangle$" (the value of x as a string) as input to the model and extracted the hidden representations from every layer $\ell \in [L]$. In cases when x is tokenized into multiple tokens, we take the representation at the last position (we assume that M is an auto-regressive model). For each basis b, we randomly split the numbers into train and validation sets with 1800 and 200 numbers, respectively.

Models We analyze two popular auto-regressive decoder-only LLMs: Llama 3 8B (Dubey et al., 2024) and Mistral 7B (Jiang et al., 2023). Llama's tokenizer contains individual tokens for all numbers between 0 and 999 inclusive, which is the common choice for modern LLMs (e.g., GPT-4 Singh and Strouse (2024) and Claude Sonnet 3.5). Mistral 7B was picked for having a different tokenization from Llama, specifically a single token per digit, which can be expected to impose a stronger bias towards digit-wise representations of numbers.

3.2 Probing recovers digit values in base 10 but not the whole number value

Table 2 shows the average probe accuracy over layers ≥ 3 in predicting all the digits of the number correctly (maximum accuracy results show similar trends; see §A.3). We do not consider the early layers as multi-token numbers require multiple layers to contextualize (see Figure 7 in §A.3).

The highest accuracy of 0.91 for Llama 3 8B and 0.92 for Mistral 7B is achieved when reconstructing the numbers in base 10. Moreover, for all other bases, accuracy is substantially lower, typically not exceeding 0.2, serving as a natural baseline for the base 10 results. Specifically, classifying the number directly (base 2000) succeeds in only $< 1\%$ of the cases, which further shows that the direct circular representation of the value in the hidden space is not accurate enough for arithmetic, similarly to the linear representation mentioned earlier. Interestingly, base 5 also has relatively high

[1] atan2 computes the two argument arctangent, which we convert from a signed to an unsigned angle between 0 and 2π.